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MCY 612
Lab 1 IAM
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MCY 612 Cloud Security
Lab#1: AWS Identity and Access Management

In this lab, you will study the Amazon Identity and Access Management (IAM) and IAM Access Analyzer. You will use the AWS CLI to implement three security best practices with IAM and use IAM Access Analyzer to identify potential resource-access risks.

Part 1: Security Best Practices in IAM

Practice 1: Creating individual IAM users

1.Don't use your AWS account root user credentials to access AWS, and don't give your credentials to anyone else. Instead, create individual users for anyone who needs access to your AWS account. Create an IAM user for yourself as well, give that user administrative permissions, and use that IAM user for all your work. In MCY 611, you have learned how to create an IAM user via the AWS management console and how to configure your AWS CloudShell. Now you will learn how to create an IAM user via AWS CLI commands. Start your AWS CloudShell terminal and then create an IAM administrative user via the following command:

aws iam create-user --user-name *your_nku_username*-admin-user

What are your command and its complete output?

```
AWS CloudShell
~ $ aws iam create-user --user-name carrd12-admin-user
{
    "User": {
        "Path": "/",
        "UserName": "carrd12-admin-user",
        "UserId": "AIDA42PHHZCTSJOBGHHJW",
        "Arn": "arn:aws:iam::881490118823:user/carrd12-admin-user",
        "CreateDate": "2025-03-23T22:28:22+00:00"
    }
}
~ $
```

Practice 2: Use user groups to assign permissions to IAM users

2.Instead of defining permissions for individual IAM users, it's usually more convenient to create user groups that relate to job functions (administrators, developers, accounting, etc.). Next, define the relevant permissions for each user group. Finally, assign IAM users to those user groups. All the users in an IAM user group inherit the permissions assigned to the user group. That way, you can make changes for everyone in a user group in just one place. As people move around in your company, you can simply change what IAM user group their IAM user belongs to. Create an IAM admin user group via the following command:

aws iam create-group --group-name *your_nku_username*-admin-group

What are your command and its complete output?

```
~ $ aws iam create-group --group-name carrd12-admin-group
{
    "Group": {
        "Path": "/",
        "GroupName": "carrd12-admin-group",
        "GroupId": "AGPA42PHHZCT3JEDWHQJK",
        "Arn": "arn:aws:iam::881490118823:group/carrd12-admin-group",
        "CreateDate": "2025-03-23T22:29:22+00:00"
    }
}
~ $
```

3.Attach the AdministratorAccess policy to our group. AdministratorAccess provides full access to AWS services and resources, so we can do almost anything that a root account can with it. However, we can maintain our root account separately in case an admin account becomes compromised. The following command attaches the AdministratorAccess policy to the admin group:

aws iam attach-group-policy --group-name *your_nku_username*-admin-group --policy-arn arn:aws:iam::aws:policy/AdministratorAccess

What is your command?

```
~ $ aws iam attach-group-policy --group-name carrd12-admin-group  --policy-arn arn:aws:iam::aws:policy/AdministratorAccess
```

4.Add the admin user to the admin group via the following command:

aws iam add-user-to-group --group-name *your_nku_username*-admin-group --user-name *your_nku_username*-admin-user

What is your command?

```
~ $ aws iam add-user-to-group --group-name carrd12-admin-group --user-name carrd12-admin-user
```

5.Creates an AWS secret access key and corresponding AWS access key ID for the admin user via the following command:

aws iam create-access-key --user-name *your_nku_username*-admin-user

What are your command and its complete output? What is your AccessKeyId? What is your SecretAccessKey?

```
~ $ aws iam create-access-key --user-name carrd12-admin-user
{
    "AccessKey": {
        "UserName": "carrd12-admin-user",
        "AccessKeyId": "AKIA42PHHZCTQDSFLZO2",
        "Status": "Active",
        "SecretAccessKey": "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",
        "CreateDate": "2025-03-23T22:32:40+00:00"
    }
}
~ $
```

6.Switch to the newly created admin account by running aws configure and supply your AccessKeyId and your SecretAccessKey.

In the terminal type: aws configure
You will be prompted for the following info:
AWS Access Key ID [****************]: *your AccessKeyId*
AWS Secret Access Key [****************]: *your SecretAccessKey*
Default region name [us-east-1]: us-east-1
Default output format [json]: json

```
~ $ aws configure
AWS Access Key ID [None]: AKIA42PHHZCTQDSFLZO2
AWS Secret Access Key [None]: nIEmYtuGuajkQujz5oNmanYMgvLRCFr/eF723fZK
Default region name [None]: us-east-1
Default output format [None]: json
~ $ 
```

Practice 3: Granting Least Privilege

7.When you create IAM policies, follow the standard security advice of granting least privilege, or granting only the permissions required to perform a task. Determine what users (and roles) need to do and then craft policies that allow them to perform only those tasks. Create a user named "*your_nku_username*-s3-user".

What are your command and its complete output?

```
~ $ aws iam create-user --user-name carrd12-s3-user
{
    "User": {
        "Path": "/",
        "UserName": "carrd12-s3-user",
        "UserId": "AIDA42PHHZCT5QMSLVQLG",
        "Arn": "arn:aws:iam::881490118823:user/carrd12-s3-user",
        "CreateDate": "2025-03-23T22:53:09+00:00"
    }
}
~ $ 
```

8.Create access keys for the newly created s3 user.

What are your command and its complete output? What is your AccessKeyId? What is your SecretAccessKey?

```
~ $ aws iam create-access-key --user-name carrd12-s3-user
{
    "AccessKey": {
        "UserName": "carrd12-s3-user",
        "AccessKeyId": "AKIA42PHHZCT2MULUXHP",
        "Status": "Active",
        "SecretAccessKey": "UkUpP5eStkOT6KoIseT4cDB5xP3+KDjGyThsnXPn",
        "CreateDate": "2025-03-23T23:03:39+00:00"
    }
}
```

9.Create a group named “*your_nku_username*-s3-group” for the s3 user.

What are your command and its complete output?

```
~ $ aws iam create-group --group-name carrd12-s3-group
{
    "Group": {
        "Path": "/",
        "GroupName": "carrd12-s3-group",
        "GroupId": "AGPA42PHHZCT7FPC3LU4W",
        "Arn": "arn:aws:iam::881490118823:group/carrd12-s3-group",
        "CreateDate": "2025-03-23T22:56:43+00:00"
    }
}
```

10.Attach a policy that limits access to S3 resources to your s3-group via the following command:

aws iam attach-group-policy --group-name *your_nku_username*-s3-group --policy-arn arn:aws:iam::aws:policy/AmazonS3FullAccess

What is your command?

```
~ $ aws iam attach-group-policy --group-name carrd12-s3-group --policy-arn arn:aws:iam::aws:policy/AmazonS3FullAccess
```

11.Add your s3 user to the s3 group.

What is your command?

```
~ $ aws iam add-user-to-group --group-name carrd12-s3-group --user-name carrd12-s3-user
```

12.Switch to the s3user account by running aws configure and supply the AccessKeyId and the SecretAccessKey that you got at step 8.

What are your command and its complete output?

```
~ $ aws configure
AWS Access Key ID [****************LZO2]: AKIA42PHHZCT2MULUXHP
AWS Secret Access Key [****************3fZK]: UkUpP5eStkOT6KoIseT4cDB5xP3+KDjGyThsnXPn
Default region name [us-east-1]: us-east-1
Default output format [json]: json
~ $
```

13.Make an s3 bucket via the following command:

aws s3 mb s3://*your_nku_username*-612

What are your command and its complete output?

```
~ $ aws s3 mb s3://carrd12-612
make_bucket: carrd12-612
```

List ec2 instances via the following command:

aws ec2 describe-instances

What is the complete output? Please explain why you got this output?

```
~ $ aws ec2 describe-instances

An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: You are not authorized to perform th
is operation. User: arn:aws:iam::881490118823:user/carrd12-s3-user is not authorized to perform: ec2:DescribeInstances becau
se no identity-based policy allows the ec2:DescribeInstances action
```

When an IAM user or role does not have the required rights to carry out the requested operation, such as describing EC2 instances, the UnauthorizedOperation issue in AWS usually occurs. The Describe Instances permission is required for the user.

Part 2: IAM Access Analyzer
Before starting this section, switch back to *your_nku_username*-admin-user by typing aws configure in the terminal and providing the keys that were created at step 5.

14.AWS IAM Access Analyzer helps identify potential resource-access risks by identifying policies that grant external access. You will use the access analyzer to identify s3 access risk later. Create an analyzer with the following command:

aws accessanalyzer create-analyzer --analyzer-name testAnalyzer --type ACCOUNT

What is the complete output? What is the ARN of your analyzer?
arn:aws:access-analyzer:us-east-1:881490118823:analyzer/testAnalyzer

```
~ $ aws accessanalyzer create-analyzer --analyzer-name testAnalyzer --type ACCOUNT
{
    "arn": "arn:aws:access-analyzer:us-east-1:881490118823:analyzer/testAnalyzer"
}
```

15.Make the bucket you created at step 13 insecure via the following steps:

Delete public access block.

aws s3api delete-public-access-block --bucket *your_nku_username*-612

Set bucket policy to allow public read access. First, create a file called policy.json with the following contents:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::your_nku_username-612/*"
            ]
        }
    ]
}
```

Set the bucket policy with this command:

aws s3api put-bucket-policy --bucket *your_nku_username*-612 --policy file://policy.json

```
~ $ aws s3api put-bucket-policy --bucket carrd12-612 --policy file://policy.json
```

16.Start a scan of the policies applied to your s3 bucket via the following command:

aws accessanalyzer start-resource-scan --analyzer-arn *your_analyzer_arn* --resource-arn arn:aws:s3:::*your_nku_username*-612

What is your command?

```
~ $ aws accessanalyzer start-resource-scan --analyzer-arn arn:aws:access-analyzer:us-east-1:881490118823:analyzer/te
stAnalyzer --resource-arn arn:aws:s3:::carrd12-612
```

17.Retrieve a list of findings generated by the analyzer via the following command:

aws accessanalyzer list-findings --analyzer-arn *your_analyzer_arn*

What is the complete output? Did the analyzer discover your bucket permission issue? You should see the listing for your bucket has the status "ACTIVE".

```
~ $ aws accessanalyzer list-findings --analyzer-arn arn:aws:access-analyzer:us-east-1:881490118823:analyzer/testAnalyzer
{
    "findings": [
        {
            "id": "6f1d4688-b70c-435b-816f-19c04cb6dcbd",
            "principal": {
                "AWS": "*"
            },
            "action": [
                "s3:GetObject"
            ],
            "resource": "arn:aws:s3:::carrd12-612",
            "isPublic": true,
            "resourceType": "AWS::S3::Bucket",
            "condition": {},
            "createdAt": "2025-03-24T01:03:19.470000+00:00",
            "analyzedAt": "2025-03-24T01:03:33.391000+00:00",
            "updatedAt": "2025-03-24T01:03:19.470000+00:00",
            "status": "ACTIVE",
            "resourceOwnerAccount": "881490118823",
            "sources": [
                {
                    "type": "POLICY"
                }
            ],
            "resourceControlPolicyRestriction": "NOT_APPLICABLE"
        }
    ]
}
```

18.Set your bucket to private via the following command:

aws s3api put-public-access-block --bucket *your_nku_username*-612 --public-access-block-configuration BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=true,RestrictPublicBuckets=true

19.Run another scan to allow the access analyzer to update its findings.

What is your command?
aws accessanalyzer start-resource-scan --analyzer-arn arn:aws:access-analyzer:us-east-1:881490118823:analyzer/testAnalyzer --resource-arn arn:aws:s3:::carrd12-612

```
~ $ aws accessanalyzer start-resource-scan --analyzer-arn arn:aws:access-analyzer:us-east-1:881490118823:analyzer/te
stAnalyzer --resource-arn arn:aws:s3:::carrd12-612
```

20.List the findings again.

What is your command and its complete output? Did you see that the listing for your bucket now has the status "RESOLVED"?

```
~ $ aws accessanalyzer list-findings --analyzer-arn arn:aws:access-analyzer:us-east-1:881490118823:analyzer/testAnalyzer
{
    "findings": [
        {
            "id": "6f1d4688-b70c-435b-816f-19c04cb6dcbd",
            "principal": {
                "AWS": "*"
            },
            "action": [
                "s3:GetObject"
            ],
            "resource": "arn:aws:s3:::carrd12-612",
            "isPublic": true,
            "resourceType": "AWS::S3::Bucket",
            "condition": {},
            "createdAt": "2025-03-24T01:03:19.470000+00:00",
            "analyzedAt": "2025-03-24T01:07:53.971000+00:00",
            "updatedAt": "2025-03-24T01:07:53.971000+00:00",
            "status": "RESOLVED",
            "resourceOwnerAccount": "881490118823",
            "sources": [
                {
                    "type": "POLICY"
                }
            ],
            "resourceControlPolicyRestriction": "NOT_APPLICABLE"
        }
    ]
}
```

Part 3: Cleanup
21.Delete the access key via the following command:

aws iam delete-access-key --access-key-id *access-key-id_of_your_nku_username-s3-user* --user-name *your_nku_username*-s3-user

You can find *access-key-id_of_your_nku_username-s3-user* at step 8. **What is your command?**

```
~ $ aws iam delete-access-key --access-key-id AKIA42PHHZCT2MULUXHP --user-name carrd12-s3-user
```

22.Remove the IAM user from the group via the following command:

aws iam remove-user-from-group --user-name *your_nku_username*-s3-user --group-name *your_nku_username*-s3-group

What is your command?

```
~ $ aws iam remove-user-from-group --user-name carrd12-s3-user --group-name carrd12-s3-group
```

23.Delete the IAM user via the following command:

aws iam delete-user --user-name *your_nku_username*-s3-user

What is your command?

```
~ $ aws iam delete-user --user-name carrd12-s3-user
```

24.Detach the group policy via the following command:

aws iam detach-group-policy --group-name *your_nku_username*-s3-group --policy-arn arn:aws:iam::aws:policy/AmazonS3FullAccess

What is your command?

```
~ $ aws iam detach-group-policy --group-name carrd12-s3-group --policy-arn arn:aws:iam::aws:policy/AmazonS3FullA
ccess
```

25.Delete the group via the following command:

aws iam delete-group --group-name *your_nku_username*-s3-group

What is your command?

```
~ $ aws iam delete-group --group-name carrd12-s3-group
```

26.Remove your S3 bucket.

What is your command and its complete output?

```
~ $ aws s3 rb s3://carrd12-612
remove_bucket: carrd12-612
```

Submission instruction
After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.

I investigated Amazon IAM and IAM Access Analyzer in this lab. I used the AWS CLI to apply security best practices, such as granting least-privilege access to IAM users and turning on multi-factor authentication. In order to make sure that AWS resources are adequately protected and reduce the possibility of unwanted access, I also used IAM Access Analyzer to find possible resource-access concerns. My comprehension of IAM's function in access control as well as how to efficiently evaluate and reduce security threats has improved as a result of this procedure.